

Software Testing & Quality Assurance

Lab Class Report: Test Case Design

Project Title: Improved Data Leakage Detection System
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Technology Stack: MERN Stack (MongoDB, Express.js, React.js, Node.js)

1. System Under Test: Improved Data Leakage Detection System

The Improved Data Leakage Detection System is a secure enterprise file-sharing platform designed to provide traceability and accountability. The system allows administrators to upload confidential files, generates unique recipient-specific copies using steganographic tracking tokens, and enables forensic leak analysis to attribute compromised files to their original recipients.

2. Part A — Equivalence Partitioning (EP)

2.1 Requirement R1 — Steganographic Tracking Token (UUID v4) Format [FR-3]

Backend generates and validates an RFC 4122 compliant UUID v4 tracking token consisting of exactly 36 characters in canonical hexadecimal form (8-4-4-4-12).

Designed Test Case (Detailed Specification)

Field	Details
Test Case ID	TC-DLD-EP-02
Test Scenario	Verify that the steganographic engine validates and accepts a canonical 36-character UUID v4 tracking token before embedding (FR-3).
Preconditions	- Backend steganography pipeline receives file allocation request. - Cover file <code>blueprint.bmp</code> is staged in storage.
Test Steps	- Invoke <code>stegoEngine.validateAndEmbed(coverFile, trackingToken)</code>. - Supply valid 36-char token: <code>c4a7e810-7b2a-4f51-9e23-8d6f1234abcd</code>. - Execute watermark embedding routine.
Input Data	- Cover File: <code>blueprint.bmp</code> - Tracking Token: <code>c4a7e810-7b2a-4f51-9e23-8d6f1234abcd</code> (Length = 36)
Expected Result	Token passes regex validation, watermark is embedded, and unique recipient copy is generated.
Actual Result	(To be completed during test execution)
Status	(Pass / Fail)
Remarks	Validates valid partition P1 (Canonical UUID) using Equivalence Partitioning (EP).

3. Part B — Boundary Value Analysis (BVA)

3.1 Requirement R2 — Minimum Cover Image Dimensions for LSB Watermarking [FR-3 / NFR-8]

For .jpg and .bmp files, the LSB steganography engine requires a minimum resolution of 256×256 pixels ($65,536px$) to embed the 128-bit encrypted token across redundant color channels without perceptible distortion.

Designed Test Case (Detailed Specification)

Field	Details
Test Case ID	TC-DLD-BVA-03
Test Scenario	Verify that the LSB steganographic engine rejects images with dimensions below 256×256 pixels using Boundary Value Analysis (FR-3 / NFR-8).
Preconditions	1. Administrator uploads image <code>icon_small.bmp</code> with resolution 255×255 pixels. 2. Administrator attempts to allocate this image to an agent.
Test Steps	1. Upload <code>icon_small.bmp</code> (255×255). 2. Select target agent. 3. Click “Generate Watermarked Copy”.
Input Data	• Filename: <code>icon_small.bmp</code> • Dimensions: 255×255 pixels
Expected Result	System blocks embedding and displays error: <i>“Steganographic error: Cover image dimensions (255x255) are below the minimum required resolution of 256x256 pixels.”</i>
Actual Result	(To be completed during test execution)
Status	(Pass / Fail)
Remarks	Validates lower boundary value $256 - 1 = 255$ pixels using Boundary Value Analysis (BVA).

3.2 Requirement R3 — Steganographic Cover Text Payload Capacity [FR-3]

For .txt files, the zero-width steganographic embedding engine requires a minimum of **50 words** in the cover document to reliably embed the unique tracking footprint without line overflow. Files with fewer than 50 words cannot accommodate the watermark.

Designed Test Case (Detailed Specification)

Field	Details
Test Case ID	TC-DLD-BVA-05
Test Scenario	Verify that the steganographic engine permits watermark embedding at exactly 50 words and blocks files with 49 words (FR-3 / BVA).
Preconditions	1. Administrator selects <code>memo_short.txt</code> containing exactly 49 words. 2. Administrator selects 1 recipient agent for distribution.
Test Steps	1. Upload <code>memo_short.txt</code>. 2. Select agent Hojaifa (24524203181). 3. Click “Generate Watermarked Copy”.
Input Data	• File: <code>memo_short.txt</code> • File Content Word Count: 49 words
Expected Result	System steganography engine returns an error: <i>“Steganographic embedding error: Cover text contains 49 words, but a minimum of 50 words is required to encode the tracking token.”</i>
Actual Result	(To be completed during test execution)
Status	(Pass / Fail)
Remarks	Validates lower boundary value $50 - 1 = 49$ words using Boundary Value Analysis (BVA).